

Structure:

It is a massive push of capital to alter and improve the power system of South Africa out to 2050. The summary is that they are going to eventually retire coal plants that have been running and being the bedrock of the power system over a period of time, and in that time they aim for natural gas plants to replace it, along with a long of wind and solar being built, along with the transmission lines that enable those to thrive. This is what the current fleet is for Eskom. Here is what the current fleet is from IPPs and the programs they have had. We program demand to reach this. Here is the committed gen from eskom. Here is the committed gen from private investment. Here is the forecasted EAF of the coal plants. Along with this, we have

- The IRP's objective is to plan for an adequate power system that will satisfy electricity demand while enduring contingencies or disruptions without adverse interruptions of supply. This is carefully crafted in such a manner that both the cost of electricity to the economy and environmental (global and local emissions) impact of electricity supply are minimized.
- Key outcomes of the irp 2025
 - Energy availability factor of 66% to 68% from 2025 to 2030 makes it possible to meet the energy demand forecast of 255 TWh in 2030
 - The quantum of renewable energy technologies increases from about 17 GW to more than 45 GW by 2030, translating to 5 GW per annum over this period
 - The minimum annual capacity factor of the 6 GW CCGT plant is 51% from 2030 to 2040
- Energy security of supply in the IRP 2025 hinges on the following critical levers
 - Sustaining the power system plant performance above 60% in the immediate to long term
 - Commissioning 6 GW of CCGT capacity by 2030 to avoid supply shortfalls as the older baseload power stations reach end of life
 - Roll out committed generation capacity, including renewables and flexible resources such as peaking gas and storage
 - Implementation of long lead time generation resources such as nuclear and water pumped storage
 - Implementation of transmission grid infrastructure as per the Transmission Development Plan (TDP)
- The IRP 2025 presents a R2.23-trillion investment plan that defines South Africa's energy mix to drive the government's deliberate effort to grow the economy for the future.

- The significance of the IRP 2025 is, particularly in this context, in its incorporation of transmission infrastructure and geographical location of supply capacity.
- The Electricity Regulation Amendment Act 2024, introduced provisions for an open market platform
- The IRP seeks to balance supply and demand in the medium to long term. In its development, the process collates planning assumptions, formulates and models scenarios, determines a reference plan derived from least cost principles subject to specified constraints and relevant government objectives.
- In the short-to-medium term, the IRP's energy planning methodology involves assessing the balance between supply and demand on an hourly chronological basis, considering limitations on the transmission network. This is done to assess the information on the operational, capacity, and energy adequacy of the power system security of supply. This exercise is conducted using a stochastic simulation technique (Monte Carlo) with Plexos simulation tool on an hourly unit commitment and economic dispatch, which does an optimization under uncertainties of demand, variability of renewable generation production (wind and solar) and unplanned outages of dispatchable resources.
- In the medium to long term, the IRP uses linear optimization algorithm in the Plexos simulation tool to determine the optimal mix of new generation capacity and transmission expansion for the planning horizon up to 2050, subject to constraints. The objective is to minimize the total system cost (NPV), which include capital, fixed and operating costs. Given the long planning horizon, the model uses chronological sampling approach to reduce the complexity of the mathematical problem while retaining as much detail.
- The representation of the transmission detail in the model is based on a linearized Direct Current Optimal Power Flow formulation, which essentially approximates it into an electricity transport model.
- The IRP 2025 electricity forecasting methodology is detailed in a full report accompanying this publication.
- Future forecasts are developed considering projections on economic indicators such as GDP, population, growth, behind the meter developments, sectoral shifts, demand management initiatives, and policy changes.
- FIGURE 2 showing electricity demand forecast and sensitivities
- The moderate GDP growth serves as baseline for the IRP 2025, reflecting a steady growth rate of 1.5% to 2% in the long term.
- FIGURE 3 showing Eskom coal fleet shutdown based on 50 year life
- Coal fired power stations form the backbone of South Africa's energy supply, with an installed capacity of approximately 42 GW. The dominance is largely attributed to the country's abundance of coal reserves.

- It enabled energy intensive industries such as mining and manufacturing, that require continuous and stable electricity to operate efficiently.
- The majority of these power stations are approaching the end of their operational life.
- Under Eskom's Generation Continued Operations (GCO) shutdown plan, the phased 50-year end life of the coal fleet will start in 2029, and continue over several years. This plan indicates a significant reduction of 8 GW in coal capacity between 2029 and 2030, followed by a further decline of 15 GW between 2034 and 2042.
- Koeberg was designed with an operational lifespan of 40 years, originally expected to shut down by 2025. However, in order to keep nuclear power in the energy mix going into the future, the design life of Koeberg was extended by 20 years between 2045 and 2046.
- There is currently 0.6 GW of installed hydro capacity in South Africa. Although the country is semi-arid, there are years with flood risk. During these periods of increased flood risk, hydropower stations are used to produce baseload energy thereby preventing the dams overflowing. During low water levels, power plants face challenges as priority is given to downstream projects water users. In such circumstances, hydropower stations can be utilized for short-duration operation to stabilize the grid if needed for emergencies.
- Although South Africa is a semi-arid country, some of its neighbours are endowed with water resources. Currently a bilateral agreement exists with Mozambique to allow import of power from their Hydro Cahora Bassa with an allocated capacity of 1 GW. There's potential to partner with Lesotho on the development of a water pumped storage scheme, Kobong, as part of the Lesotho Highlands Water Project.
- South Africa is one of the most endowed countries for solar and wind generation due to the country's higher than average wind speeds and solar radiation. The country experiences some of the highest levels of solar radiation globally along the northwestern regions, while strong and consistent wind patterns are prevalent in the Cape areas.
- South Africa's public procurement processes for the electricity sector is espoused in the National Development Plan 2030.
- Independent Power Producers Procurement Program (IPPPP), the total capacity already developed is 6.2 GW, procured from Renewable Energy IPP.
- FIGURE 4 COMMITTED PUBLIC GENERATING CAPACITY
- FIGURE 5 COMMITTED PRIVATE GENERATING CAPACITY
- Development of private generation projects is enabled by the amended Schedule 3 of the Electricity Regulation Act 4 of 2006, which allows for embedded generation projects to be developed without needing a license.

- Private projects in the pipeline, with all permitting and licensing secured, have also advanced in securing grid capacity allocation, totaling 13 GW as shown in figure 5. Those already operational, are under construction or have an issued budget quotation. Make up approximately 55% or 8 GW. The remaining 45% are at the Intertim Grid Capacity Allocation Rules (IGCAR) stage.
- **FIGURE 6 COMMITTED ESKOM GENERATION CAPACITY**
- The period up to 2030 is characterized by committed new generation resources, driven primarily by independent power producers participation. To maintain power system stability, this period is projected to be sufficiently enabled by existing policy and initiatives already in place. Key amongst these are:
 - Sustain progress made on the Eskom Recovery Plan as directed by the energy action plan
 - Ensure committed generation projects (public, private, Eskom) are executed expeditiously
 - Advance the gas to power programme
 - Advance the independent transmission project initiative to allow investment in expanding and strengthening the electricity infrastructure
- Three of these five pillars are relevant in the context of the current IRP review:
 - Fixing Eskom's existing power stations plant performance (EAF)
 - Enable and accelerate private investment in new generation capacity
 - Encourage and fast track distributed generation
- Restoration of ferrochrome industry
- The requirement for gas CCGT in South African power system by 2030 is critical in lieu of the impending decommissioning of 8 GW of baseload coal fired plant
- The gas CCGT is committed by 2030, then an additional 1 GW of gas-to-power new generation capacity commissioned per annum is required between 2030 and 2040.
- Gas Master Plan
- Development of electricity transmission infrastructure in synchrony with the delivery of gas molecules and completion of the generating plant
- Securing LNG and associated infrastructure in South Africa by 2030 remains a risk and may require policy adjustment to ensure timelines are met
- South Africa must develop a Nuclear Industrialization Plan
- Eskom intends to shut down the 8 GW of coal fired power stations, including Camden, Hendrina, Grootvlei, and Arnot by financial year 2030.
- Just Energy Transition strategy to guide the phased shut down of coal fired power stations up to 2030
- This approach is anchored on the principles of SA's JET framework.

- After careful consideration of the observations made from the technical analysis done to produce the Proposed Balanced Plan, the following policy statements can be made with a high degree of certainty to ensure a resilient power system
 - The EAF of the current Eskom fleet to be maintained at levels above 60%. Any levels below 60% will result in a fragile power system that will not only be able to absorb negative contingencies but will also not be able to support aspirations to grow the economy
 - The 6 GW of gas CCGT by 2030 is crucial in ensuring and maintaining security of supply when the 8 GW of baseload operation of coal fired stations is shutdown
 - Demonstration plant on clean coal tech by 2030 as the country explores other options which are potentially cheaper and efficient to significantly reduce both global and local emissions. May lead to operating coal plant beyond the 50 year life of the plant.
 - Develop a Nuclear Industrialization Plant
 - Ensure effective implementation and monitoring of the South African Renewable Energy Industrialization MasterPlan
 - Consider implementation of transformative opportunity for mega bid window